Topic 10: Space

Mathematical skills that could be developed in this topic include using approximations and sketching relationships which are modelled by $y = k/x^2$.

This topic may be studied using contexts such as the formation and evolution of stars and the history and future of the universe.

Students should:
156. be able to use the equation, intensity $I = \frac{L}{4\pi d^2}$ where L is luminosity and d is distance from the source
157. understand how astronomical distances can be determined using trigonometric parallax
158. understand how astronomical distances can be determined using measurements of intensity received from standard candles (objects of known luminosity)
159. be able to sketch and interpret a simple Hertzsprung-Russell diagram that relates stellar luminosity to surface temperature
160. understand how to relate the Hertzsprung-Russell diagram to the life cycle of stars
161. understand how the movement of a source of waves relative to an observer/detector gives rise to a shift in frequency (Doppler effect)
162. be able to use the equations for redshift $z = \frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$ for a source of electromagnetic radiation moving relative to an observer and $v = H_0 d$ for objects at cosmological distances
163. understand the controversy over the age and ultimate fate of the universe associated with the value of the Hubble constant and the possible existence of dark matter.